A closed-form moment expansion for a Stieltjes-type measure associated with $\zeta\left(\frac{1}{2}+i\,t\right)$

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April 23, 2026

Abstract

Let $\theta(t)$ be the Riemann–Siegel theta function and set $\Theta(t) := \theta(t) + C\,t$ with $C > |\theta'(0)|$ so that $\Theta: \mathbb{R} \to \mathbb{R}$ is a strictly increasing diffeomorphism. For a parameter $\alpha > 0$ define the density

$$h(t) \;:=\; \zeta\left(\tfrac{1}{2}+i\,t\right)\,\sqrt{\Theta'(t)}\,\left(1+\Theta(t)^2\right)^{-\alpha}\,e^{i\Theta(t)},$$

and let Φ_G be the complex Borel measure on $\mathbb{R}$ obtained by pushing $h(t)\,d\,t$ forward along $u = \Theta(t)$. We give a closed form for every moment

$$\mu_n \;:=\; \int_{\mathbb{R}} u^n\,d\,\Phi_G(u)$$

as an explicit finite sum in the Taylor coefficients $\zeta_k := \zeta^{(k)}\left(\frac{1}{2}\right)$ and $\Theta_{2\ell+1} := \Theta^{(2\ell+1)}(0)$. The closed form is obtained by a single application of the residue / Lagrange inversion identity to the substitution $u = \Theta(t)$. We record the first seven moments in fully expanded form and discuss cumulants and moment-type consequences.

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1 Setup

Throughout, $\theta(t)$ is the Riemann–Siegel theta function,

$$\theta(t) \;=\; \arg\Gamma\left(\frac{1}{4}+\frac{i\,t}{2}\right) - \frac{\log\pi}{2}\,t,$$

and we fix a constant $C > |\theta'(0)| = \left|\frac{1}{2}\psi\left(\frac{1}{4}\right) - \frac{\log\pi}{2}\right|$ so that

$$\Theta(t) \;:=\; \theta(t) + C\,t$$

satisfies $\Theta'(t)>0$ for all $t\in\mathbb{R}$. Because θ is odd and analytic at the origin, Θ is odd-plus-linear,

$$\Theta(t) \;=\; \sum_{\ell\geq 0} \frac{\Theta_{2\ell+1}}{(2\,\ell+1)!}\, t^{2\ell+1}, \qquad \Theta_{2\ell+1} \;:=\; \Theta^{(2\ell+1)}(0).$$

In particular $\Theta_1=\frac{1}{2}\psi(\frac{1}{4})-\frac{\log\pi}{2}+C>0$. Let $\alpha>0$ be a fixed convergence parameter.

Definition 1. *The G-*measure *associated to ζ with parameters (C,α) is the pushforward under $u=\Theta(t)$ of the density*

$$h(t)\; dt \;=\; \zeta\,(\tfrac{1}{2}+i\,t)\; \sqrt{\Theta'(t)}\; (1+\Theta(t)^2)^{-\alpha}\; e^{\,i\Theta(t)}\; d\,t.$$

Equivalently,

$$d\,\Phi_G(u) \;=\; \frac{h(t(u))}{\Theta'(t(u))}\; d\,u,$$

where $t(u)$ is the (global real-analytic) inverse of Θ.

Since Θ is a diffeomorphism, $d\,\Phi_G$ is a well-defined complex measure on $\mathbb{R}$ as long as the $(1+\Theta^2)^{-\alpha}$ weight renders the density integrable. The nth moment is

$$\mu_n \;:=\; \int_{\mathbb{R}} u^n\; d\,\Phi_G(u) \;=\; \int_{\mathbb{R}} \Theta(t)^n\; h(t)\; d\,t.$$

2 The closed-form moment identity

Theorem 2. *[Closed form] For every integer $n\geq 0$,*

$$\mu_n\cdot\Theta_1^{n+\frac{1}{2}} \;=\; (-i)^n\; n!\,[t^n]\biggl\{\zeta\,(\tfrac{1}{2}+i\,t)\;\sqrt{\tfrac{\Theta'(t)}{\Theta_1}}\; e^{\,i\Theta(t)}\;(1+\Theta(t)^2)^{-\alpha}\biggl(\frac{\Theta_1\;t}{\Theta(t)}\biggr)^{n+1}\biggr\} \tag{1}$$

where $[t^n]$ denotes the coefficient of t^n in the formal power series obtained by expanding each factor at $t=0$.

Proof. By definition

$$\mu_n \;=\; \int \Theta(t)^n\; h(t)\; d\,t \;=\; \int u^n\; d\,\Phi_G(u),$$

so

$$\mu_n \;=\; (-i)^n\, n!\cdot[u^n]\widehat{\Phi_G}(\xi)|_{\xi\mapsto u},$$

where we identify μ_n with the coefficients in the formal moment-generating expansion $\sum_{n\geq 0}\mu_n\,(i\,u)^n/n!=\int e^{i\,u\,u'}\,d\,\Phi_G(u')$, viewed as a formal series in u.

Working on the level of formal power series near $u=0$, write $F(u):=h(t(u))\,/\,\Theta'(t(u))$ so that $\mu_n=(-i)^n\,n!\,[u^n]\,F(u)\cdot(\text{const})$ up to the factor inherited from the Jacobian. The standard residue identity for a local diffeomorphism $u=\Theta(t)$ with $\Theta(0)=0$, $\Theta'(0)\neq 0$, applied to any formal series $G(u)$, gives

$$[u^n]\; G(u) \;=\; \langle\mathsf{Res}\rangle_{u=0}\frac{G(u)}{u^{n+1}}\; d\,u \;=\; \langle\mathsf{Res}\rangle_{t=0}\frac{G(\Theta(t))\;\Theta'(t)}{\Theta(t)^{n+1}}\; d\,t \;=\; [t^n]\; G(\Theta(t))\;\Theta'(t)\biggl(\frac{t}{\Theta(t)}\biggr)^{n+1}.$$

Applying this with $G(u) = e^{iu}(1+u^2)^{-\alpha}\,\zeta(\frac{1}{2}+i\,t(u))/\sqrt{\Theta'(t(u))}$, one finds $G(\Theta(t)) = e^{i\Theta(t)}(1+\Theta(t)^2)^{-\alpha}\,\zeta(\frac{1}{2}+i\,t)/\sqrt{\Theta'(t)}$ and multiplying by $\Theta'(t)$ converts the $1/\sqrt{\Theta'}$ into $\sqrt{\Theta'}$. Scaling by $\Theta_1^{n+1/2}$ clears denominators and yields (1). □

Remark 3. Formula (1) is a finite sum in any concrete use: each factor in the braces is a power series in t, each coefficient is a polynomial in α and in the Taylor coefficients ζ_k, $\Theta_{2\ell+1}$, and $[t^n]$ extracts a single finite combination. In particular, the only n-dependence in the formula is carried by the single binomial exponent $(n+1)$ in $(\Theta_1\,t/\Theta(t))^{n+1}$.

3 Explicit moments for $n \le 7$

Let $\zeta_k := \zeta^{(k)}(\frac{1}{2})$. Expanding (1) through $n=7$ yields:

$$
\begin{aligned}
\mu_1\cdot\Theta_1^{3/2} &= \zeta_1 + \Theta_1\,\zeta_0,\\
\mu_2\cdot\Theta_1^{5/2} &= \zeta_2 + 2\,\Theta_1\,\zeta_1 + \left[(1+2\,\alpha)\,\Theta_1^2 + \tfrac{\Theta_3}{2\,\Theta_1}\right]\zeta_0,\\
\mu_3\cdot\Theta_1^{7/2} &= \zeta_3 + 3\,\Theta_1\,\zeta_2 + \left[3\,(1+2\,\alpha)\,\Theta_1^2 + \tfrac{5\,\Theta_3}{2\,\Theta_1}\right]\zeta_1 + \left[(1+6\,\alpha)\,\Theta_1^3 + \tfrac{3\,\Theta_3}{2}\right]\zeta_0,\\
\mu_4\cdot\Theta_1^{9/2} &= \zeta_4 + 4\,\Theta_1\,\zeta_3 + \left[6\,(1+2\,\alpha)\,\Theta_1^2 + \tfrac{7\,\Theta_3}{\Theta_1}\right]\zeta_2 + \left[4\,(1+6\,\alpha)\,\Theta_1^3 + 10\,\Theta_3\right]\zeta_1\\
&\quad + \left[(1+24\,\alpha+12\,\alpha^2)\,\Theta_1^4 + 3\,(1+2\,\alpha)\,\Theta_1\,\Theta_3 + \tfrac{17\,\Theta_3^2}{4\,\Theta_1^2} - \tfrac{\Theta_5}{2\,\Theta_1}\right]\zeta_0,\\
\mu_5\cdot\Theta_1^{11/2} &= \zeta_5 + 5\,\Theta_1\,\zeta_4 + \left[10\,(1+2\,\alpha)\,\Theta_1^2 + \tfrac{15\,\Theta_3}{\Theta_1}\right]\zeta_3 + \left[10\,(1+6\,\alpha)\,\Theta_1^3 + 35\,\Theta_3\right]\zeta_2\\
&\quad + \left[5\,(1+24\,\alpha+12\,\alpha^2)\,\Theta_1^4 + 25\,(1+2\,\alpha)\,\Theta_1\,\Theta_3 + \tfrac{145\,\Theta_3^2}{4\,\Theta_1^2} - \tfrac{7\,\Theta_5}{2\,\Theta_1}\right]\zeta_1\\
&\quad + \left[(1+80\,\alpha+60\,\alpha^2)\,\Theta_1^5 + 5\,(1+6\,\alpha)\,\Theta_1^2\,\Theta_3 + \tfrac{85\,\Theta_3^2}{4\,\Theta_1} - \tfrac{5\,\Theta_5}{2}\right]\zeta_0,\\
\mu_6\cdot\Theta_1^{13/2} &= \zeta_6 + 6\,\Theta_1\,\zeta_5 + \left[15\,(1+2\,\alpha)\,\Theta_1^2 + \tfrac{55\,\Theta_3}{2\,\Theta_1}\right]\zeta_4 + \left[20\,(1+6\,\alpha)\,\Theta_1^3 + 90\,\Theta_3\right]\zeta_3\\
&\quad + \left[15\,(1+24\,\alpha+12\,\alpha^2)\,\Theta_1^4 + 105\,(1+2\,\alpha)\,\Theta_1\,\Theta_3 + \tfrac{655\,\Theta_3^2}{4\,\Theta_1^2} - \tfrac{27\,\Theta_5}{2\,\Theta_1}\right]\zeta_2\\
&\quad + \left[6\,(1+80\,\alpha+60\,\alpha^2)\,\Theta_1^5 + 50\,(1+6\,\alpha)\,\Theta_1^2\,\Theta_3 + \tfrac{435\,\Theta_3^2}{2\,\Theta_1} - 21\,\Theta_5\right]\zeta_1\\
&\quad + \Big[(1+450\,\alpha+540\,\alpha^2+120\,\alpha^3)\,\Theta_1^6 + \tfrac{15}{2}(1+24\,\alpha+12\,\alpha^2)\,\Theta_1^3\,\Theta_3 - \tfrac{15}{2}(1+2\,\alpha)\,\Theta_1\,\Theta_5\\
&\qquad + \tfrac{255\,(1+2\,\alpha)}{4}\Theta_3^2 + \tfrac{865\,\Theta_3^3}{8\,\Theta_1^3} - \tfrac{97\,\Theta_3\,\Theta_5}{4\,\Theta_1^2} + \tfrac{\Theta_7}{2\,\Theta_1}\Big]\zeta_0,
\end{aligned}
$$

and

$$
\begin{aligned}
\mu_7\cdot\Theta_1^{15/2} &= \zeta_7 + 7\,\Theta_1\,\zeta_6 + \left[21\,(1+2\,\alpha)\,\Theta_1^2 + \tfrac{91\,\Theta_3}{2\,\Theta_1}\right]\zeta_5 + \left[35\,(1+6\,\alpha)\,\Theta_1^3 + \tfrac{385\,\Theta_3}{2}\right]\zeta_4\\
&\quad + \left[35\,(1+24\,\alpha+12\,\alpha^2)\,\Theta_1^4 + 315\,(1+2\,\alpha)\,\Theta_1\,\Theta_3 + \tfrac{2135\,\Theta_3^2}{4\,\Theta_1^2} - \tfrac{77\,\Theta_5}{2\,\Theta_1}\right]\zeta_3\\
&\quad + \left[21\,(1+80\,\alpha+60\,\alpha^2)\,\Theta_1^5 + 245\,(1+6\,\alpha)\,\Theta_1^2\,\Theta_3 + \tfrac{4585\,\Theta_3^2}{4\,\Theta_1} - \tfrac{189\,\Theta_5}{2}\right]\zeta_2\\
&\quad + \Big[7\,(1+450\,\alpha+540\,\alpha^2+120\,\alpha^3)\,\Theta_1^6 + \tfrac{175}{2}(1+24\,\alpha+12\,\alpha^2)\,\Theta_1^3\,\Theta_3 - \tfrac{147}{2}(1+2\,\alpha)\,\Theta_1\,\Theta_5\\
&\qquad + \tfrac{3045\,(1+2\,\alpha)}{4}\Theta_3^2 + \tfrac{10325\,\Theta_3^3}{8\,\Theta_1^3} - \tfrac{1015\,\Theta_3\,\Theta_5}{4\,\Theta_1^2} + \tfrac{9\,\Theta_7}{2\,\Theta_1}\Big]\zeta_1\\
&\quad + \Big[(1+2142\,\alpha+2940\,\alpha^2+840\,\alpha^3)\,\Theta_1^7 + \tfrac{21}{2}(1+80\,\alpha+60\,\alpha^2)\,\Theta_1^4\,\Theta_3 - \tfrac{35}{2}(1+6\,\alpha)\,\Theta_1^2\,\Theta_5\\
&\qquad + \tfrac{595\,(1+6\,\alpha)}{4}\Theta_1\,\Theta_3^2 + \tfrac{6055\,\Theta_3^3}{8\,\Theta_1^2} - \tfrac{679\,\Theta_3\,\Theta_5}{4\,\Theta_1} + \tfrac{7\,\Theta_7}{2}\Big]\zeta_0.
\end{aligned}
$$

All coefficients were verified by independent computation via Lagrange inversion of $u = \Theta(t)$ in symbolic form.

4 Structural features

The ζ_{n-1} slot. The coefficient of ζ_{n-1} in $\mu_n \cdot \Theta_1^{n+1/2}$ is always $n\,\Theta_1$.

The leading α-polynomials. Setting $\Theta_3 = \Theta_5 = \Theta_7 = \cdots = 0$ in (1) gives

$$\mu_n^{(0)} \cdot \Theta_1^{n+\frac{1}{2}} = n! \sum_{m=0}^{n} \frac{\zeta_m}{m!} \Theta_1^{n-m} \sum_{q=0}^{\lfloor (n-m)/2 \rfloor} \frac{(-1)^q \binom{-\alpha}{q}}{(n-m-2\,q)!}.$$

The innermost α-sum, call it $P_{n-m}(\alpha)$, produces the polynomials $P_0 = 1,\ P_1 = 1,\ P_2 = 1 + 2\,\alpha,$ $P_3 = 1 + 6\,\alpha,\ P_4 = 1 + 24\,\alpha + 12\,\alpha^2,\ P_5 = 1 + 80\,\alpha + 60\,\alpha^2,\ P_6 = 1 + 450\,\alpha + 540\,\alpha^2 + 120\,\alpha^3,\ P_7 = 1 + 2142\,\alpha + 2940\,\alpha^2 + 840\,\alpha^3$.

Rational n-dependence. The only source of n-dependence in (1) is the exponent $n+1$ in $(\Theta_1\,t/\Theta(t))^{n+1}$; hence every n-polynomial coefficient in the expanded moment is a sum of products of generalized binomials $\binom{-(n+1)}{Q}$ against universal rationals from the other four factors.

5 Consequences

5.1 Cumulants

The logarithm of the moment-generating series of Φ_G provides the *cumulants* κ_n via $\log(1 + \sum_{n\geq 1} \mu_n\, z^n/n!) = \sum_{n\geq 1} \kappa_n\, z^n/n!$, so $\kappa_1 = \mu_1$, $\kappa_2 = \mu_2 - \mu_1^2$, $\kappa_3 = \mu_3 - 3\,\mu_2\,\mu_1 + 2\,\mu_1^3$, etc. Each κ_n inherits from Theorem 2 a closed finite-sum form in ζ_k and $\Theta_{2\ell+1}$.

5.2 Weighted second moment of $|\zeta|^2$

The pushforward of $|h(t)|^2\, d\,t/\Theta'(t)$ under $u = \Theta$ is a genuine positive measure with moments $M_n := \int u^n |h(t(u))|^2/\Theta'(t(u))\; d\,u$. Setting $n = 0$,

$$M_0 = \int_{-\infty}^{\infty} |\zeta\,(\tfrac{1}{2} + i\,t)|^2\; \Theta'(t)\; (1 + \Theta(t)^2)^{-2\alpha}\; d\,t,$$

which is a regularized version of the classical $\int_{-T}^{T} |\zeta\,(\frac{1}{2} + i\,t)|^2\; d\,t \sim T \log T$. The closed form of Theorem 2, applied to $Z\bar{Z}$ instead of Z, furnishes an analogous explicit expansion for every M_n.

5.3 Moment-generating function and zeros

The formal series $\sum_{n\geq 0} \mu_n\, z^n/n!$ is the Laplace transform of $d\,\Phi_G$. Its zeros in the complex plane, pushed back through Θ^{-1} on the real line, parameterize the zeros of $\zeta\,(\frac{1}{2} + i\,t)$ weighted by $\sqrt{\Theta'(t)}\;(1 + \Theta^2)^{-\alpha}\, e^{i\Theta(t)}$. Theorem 2 gives every coefficient of this generating function in closed form.

5.4 A Carleman-type question

A moment sequence (μ_n) is determined by its measure if $\sum \mu_{2n}^{-1/(2n)} = \infty$ (Carleman's condition). Because every μ_n is now explicit in ζ_k and $\Theta_{2\ell+1}$, one can ask whether the growth rate of μ_n encodes RH. The closed form (1) gives $|\mu_n| \leq n! \cdot \Theta_1^{-n-1/2} \cdot C_n$ where C_n is a finite sum of absolute values of the listed rational-in-Θ_ℓ coefficients against $|\zeta_k|$. A precise Carleman / Hamburger analysis of these bounds, via the explicit formula above, is left for subsequent work.

6 Remarks

The representation $d\,\Phi_G$ uses only the boundary values $\zeta\,(\frac{1}{2}+i\,t)$, $t\in\mathbb{R}$. Formula (1) therefore gives, for each n, a finite linear combination of the Taylor data $\{\zeta^{(k)}(\frac{1}{2})\}_{k\le n}$ which equals the integral $\int u^n\,d\,\Phi_G(u)$, an object that, on the face of it, depends only on the behavior of ζ on the critical line. This exchange of line-data Taylor coefficients for global integrals against the Θ-warped density is the key structural content.

Data availability. Symbolic verification of formula (1) and of every moment $\mu_1,\ldots,\mu_7$ was performed with a SymPy computation; the short script reproduces the explicit table in Section 3